Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 FT 1 HT	(a)	(i)		Unit conversion for power (i.e. 0.5) **and** time (i.e. 1.5) (1) Units used = 0.5 × 1.5 = 0.75 (1) N.B. 45 000 gets 1 mark Substitution into: cost = 0.75 **ecf** × 16 or 0.16 (1) Cost = 12 [p] or £0.12 (1) **Don't accept** £0.12 p N.B. If multiplied by 0.16 4th mark can only be awarded if answer given in p or unit changed on answer line Final answers of 720 000 or 12 000 or 720 **award 3 marks** To allow **ecf** for 3rd and 4th marks, an attempt to calculate the number of units must have been made If selected incorrect line from table with e.g. correct conversion award up to 3 marks	 1	1 1 1		 4	 4	

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
		(ii)		Washing at 30^o uses <u>33%</u> less energy or time or units than at 40^o / washing at 30^o uses <u>67%</u> of the energy or time or units compared with 40^o (1) and <u>50%</u> less energy or time or units than at 50^o (1) so the claim is not correct (1) N.B. If only one of the higher temperatures is considered then award a maximum of 2 marks if the conclusion is consistent with their answer **Alternative:** At 40^o a 40% reduction in units used = <u>0.45</u> units [compared to 0.5 units at 30^o](1) At 50^o a 40% reduction in units used = <u>0.6</u> units [compared to 0.5 units at 30^o] (1) So the claim is not correct (1) N.B. If only one of the higher temperatures is considered then award a maximum of 2 marks if the conclusion is consistent with their answer **Alternative:** At 40^o a 40% reduction in time = <u>54</u> minutes [compared to 60 minutes at 30^o](1) At 50^o a 40% reduction in time = <u>72</u> minutes [compared to 60 minutes at 30^o] (1) So the claim is not correct (1) N.B. If only one of the higher temperatures is considered then award a maximum of 2 marks if the conclusion is consistent with their answer			3	3	3	

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		**Select one pair of statements** from below: - Reducing CO_2 [emissions] or greenhouse gases (1) so less [contribution to] global warming or greenhouse effect (1) - Reducing SO_2 [emissions] (1) so less [contribution to] acid rain (1) - Less use of fossil fuels (1) so less mining required / less transport [of fuel] / less CO_2 or SO_2 [emissions] / less [contribution to] global warming or greenhouse effect / so less [contribution to] acid rain (1) - Less output from power stations / less power stations needed (1) so less fossil fuels used / less CO_2 or SO_2 [emissions] / less mining required / less transport / less radioactive waste / less [contribution to] global warming or greenhouse effect / so less [contribution to] acid rain (1) **Don't accept** less pollution / less gases / harm the environment / less energy / less electricity / no CO_2 or SO_2 [emissions]		2		2		
	(b)	(i)		Live or L	1			1		
		(ii)		If a fault develops to make a metal casing live / live wire touches the metal case (1) it takes the current safely to ground / it prevents electric shock or protects the user / causes the fuse to blow or trips the mcb or rccb (1) **Don't accept** takes the electricity to ground or reference to electrified or take away excess current	2			2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Power = 230 × 3 (1) Power = 690 [W] (1) So the fuse is not suitable since this is lower than the 2 400 W required (1) **Alternative:** Voltage = $\frac{2\,400}{3}$ (1) Voltage = 800 [V] (1) So the fuse is not suitable because the voltage is only 230 V (1) **Alternative**: Current = $\frac{2\,400}{230}$ (1) Current = 10[.4 A] (1) So fuse is not suitable as it is rated less than 10.4 A (1) **N.B. Any of the answers shown above without workings award 2 marks and with correct conclusion award 3 marks** **N.B.2 If two different calculations of power / voltage / current are seen and not qualified award 1 mark only for correct conclusion** **N.B.3 Correct conclusion resulting from incorrect calculations award 1 mark**			3	3	3	
				Question 6 FT **Question 1 HT total**	**4**	**5**	**6**	**15**	**10**	**0**